Section 4.4: Part B: Probability and the Standard Distribution

Finite Mathematics MTH 107 Fall 2014

Dr. James Ashe

Mars Hill University

Friday 7th November, 2014

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Two key ideas

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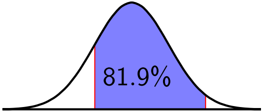

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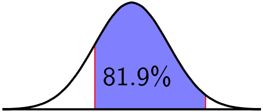

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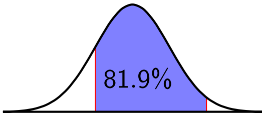

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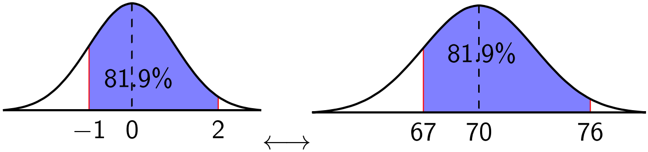

Standard Distribution: Probability

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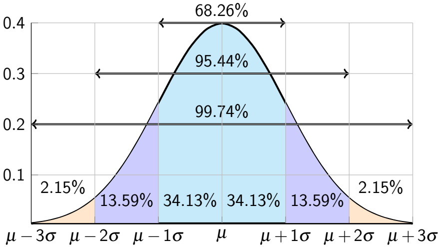

Standard Distribution

a. $68.26\% + 13.59\% = 81.85\%$
b. 81.85%
c. 50%

Normal Distribution: Probability

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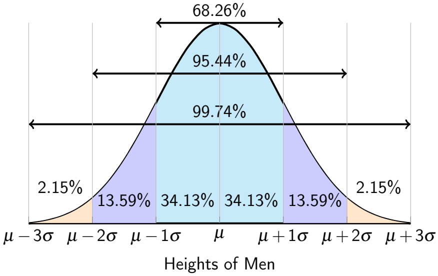

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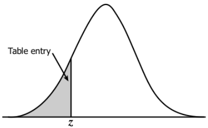

Table entry for z is the area
to the left of z.

z	.00	.01	.02	.03	.04	.05	.0
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.00
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.00
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.00
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.00
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.00
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.00
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.00
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.00

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−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.05
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.07
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.08
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.10
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.12
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.14
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.16
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.19
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.22
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.25
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.28
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.32
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.35
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.39
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.43
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.47

a. 95.05%

b. $95.05\% - 50\% = 45.05\%$

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−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.12
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3	.0778	.0764	.0749	.0735	.0721	.0708	.0694
1	.0934	.0918	.0901	.0885	.0869	.0853	.0838
1	.1112	.1093	.1075	.1056	.1038	.1020	.1003
5	.1314	.1292	.1271	.1251	.1230	.1210	.1190
2	.1539	.1515	.1492	.1469	.1446	.1423	.1401
4	.1788	.1762	.1736	.1711	.1685	.1660	.1635
0	.2061	.2033	.2005	.1977	.1949	.1922	.1894
9	.2358	.2327	.2296	.2266	.2236	.2206	.2177
9	.2676	.2643	.2611	.2578	.2546	.2514	.2483
0	.3015	.2981	.2946	.2912	.2877	.2843	.2810
9	.3372	.3336	.3300	.3264	.3228	.3192	.3156
3	.3745	.3707	.3669	.3632	.3594	.3557	.3520
8	.4129	.4090	.4052	.4013	.3974	.3936	.3897
2	.4522	.4483	.4443	.4404	.4364	.4325	.4286
0	.4920	.4880	.4840	.4801	.4761	.4721	.4681

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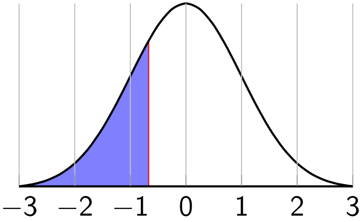

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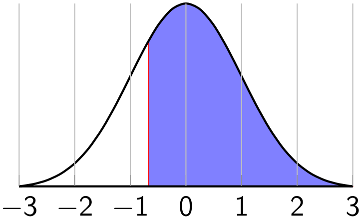

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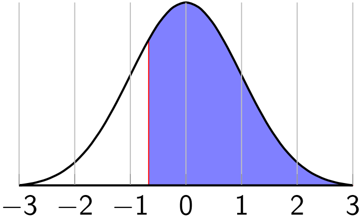

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b. $95.05\% - 50\% = 45.05\%$
c. 25.14%
d. $100\% - 25.14\% = 74.86\%$

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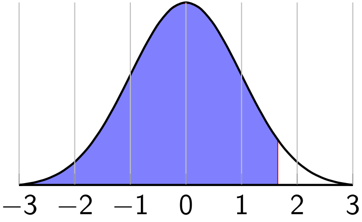

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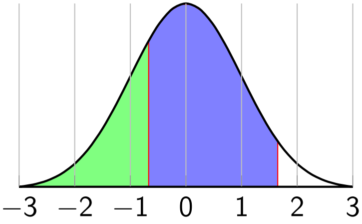

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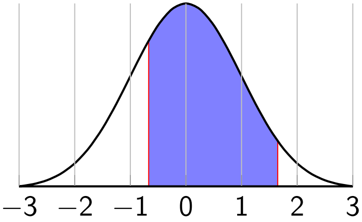

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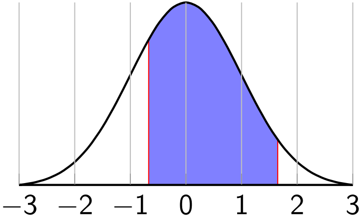

a. 95.05%
b. $95.05\% - 50\% = 45.05\%$
c. 25.14%
d. $100\% - 25.14\% = 74.86\%$
e. $95.05\% - 25.14\% = 69.91\%$

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Example 4.4.4.

Find the following probabilities.

a. $p(z < 1.42)$

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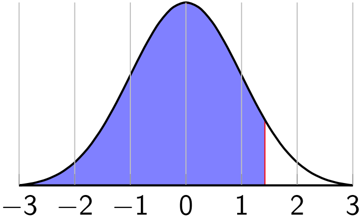
Example 4.4.4.

Find the following probabilities.

a. $p(z < 1.42)$
b. Let $\mu = 5$ and $\sigma = 10$; $p(x < 19.2)$



a. 92.22%

Standard Distribution: Probability

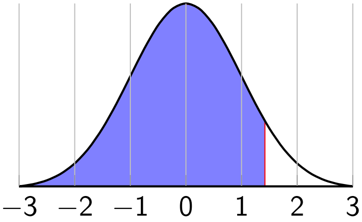
Applet: http://www.mathsisfun.com/data/standard-normal-distribution-table.html
Chart: http://www.stat.ufl.edu/ athienit/Tables/Ztable.pdf

Example 4.4.4.

Find the following probabilities.

a. $p(z < 1.42)$

b. Let $\mu = 5$ and $\sigma = 10$; $p(x < 19.2)$



a. 92.22%

b. $19.2 = 5 + 1.42 \cdot 10$

Standard Distribution: Probability

Applet: http://www.mathsisfun.com/data/standard-normal-distribution-table.html
Chart: http://www.stat.ufl.edu/ athienit/Tables/Ztable.pdf

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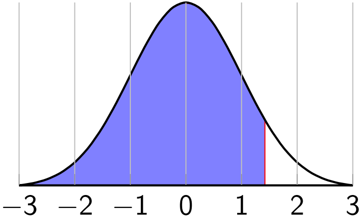

a. 92.22%

b. $19.2 = 5 + 1.42 \cdot 10$

$x = 19.2$ is 1.42 standard deviations above the mean.

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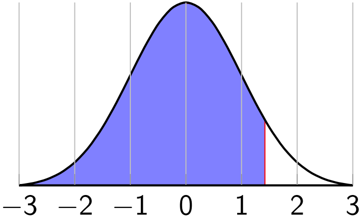

a. 92.22%
b. $19.2 = 5 + 1.42 \cdot 10$

   $x = 19.2$ is 1.42 standard deviations above the mean.

   $p(x < 19.2) = 92.22\%$

Normal Distribution: Probability

Normal Distribution: Probability

Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.

Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.

Normal Distribution: Probability

Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

a. $p(x < 19.2)$

Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.

Normal Distribution: Probability

Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

a. $p(x < 19.2)$

b. $p(15 < x)$

Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.

Normal Distribution: Probability

Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

a. $p(x < 19.2)$

b. $p(15 < x)$

c. $p(11 < x < 15)$

Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.

Normal Distribution: Probability

Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

a. $p(x < 19.2)$

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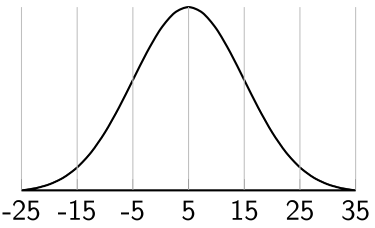


a. $p(x < 19.2) = p(z < 1.42) =$ 92.22%

b. $z = \dfrac{x-5}{10}$

Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

a. $p(x < 19.2)$

b. $p(15 < x)$

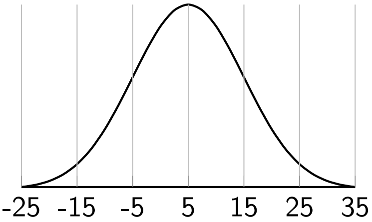
c. $p(11 < x < 15)$

Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.



a. $p(x < 19.2) = p(z < 1.42) =$ 92.22%

b. $z = \dfrac{15 - 5}{10}$

Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

a. $p(x < 19.2)$

b. $p(15 < x)$

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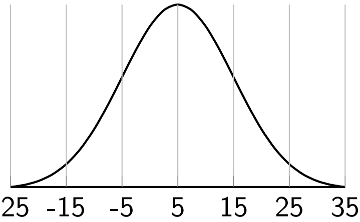
Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.



a. $p(x < 19.2) = p(z < 1.42) =$ 92.22%

b. $1 = \dfrac{15 - 5}{10}$

Normal Distribution: Probability

Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

a. $p(x < 19.2)$

b. $p(15 < x)$

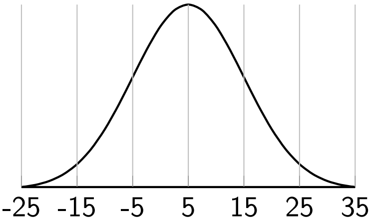
c. $p(11 < x < 15)$

Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.



a. $p(x < 19.2) = p(z < 1.42) = 92.22\%$

b. $p(x < 15) = p(z < 1) = 84.13\%$

Normal Distribution: Probability

Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

a. $p(x < 19.2)$

b. $p(15 < x)$

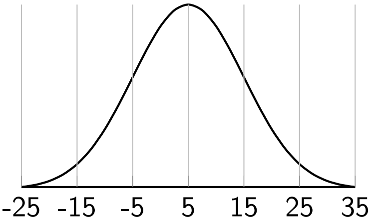
c. $p(11 < x < 15)$

Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.



a. $p(x < 19.2) = p(z < 1.42) = 92.22\%$

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Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

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Convert a Normal to a Z

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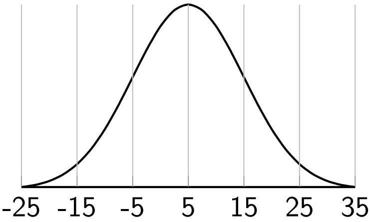

a. $p(x < 19.2) = p(z < 1.42) = 92.22\%$

b. $p(x < 15) = p(z < 1) = 84.13\%$

c. $z = \dfrac{x - 5}{10}$

Normal Distribution: Probability

Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

a. $p(x < 19.2)$

b. $p(15 < x)$

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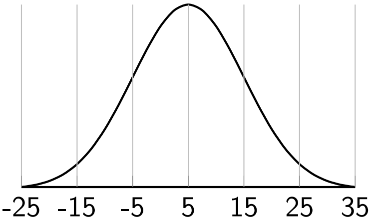
Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.



a. $p(x < 19.2) = p(z < 1.42) = 92.22\%$

b. $p(x < 15) = p(z < 1) = 84.13\%$

c. $z = \dfrac{11 - 5}{10}$

Normal Distribution: Probability

Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

a. $p(x < 19.2)$

b. $p(15 < x)$

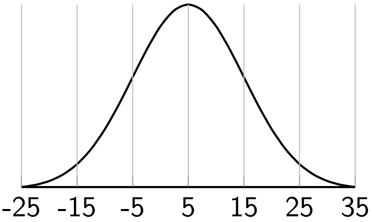
c. $p(11 < x < 15)$

Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.



a. $p(x < 19.2) = p(z < 1.42) = 92.22\%$

b. $p(x < 15) = p(z < 1) = 84.13\%$

c. $.6 = \dfrac{11 - 5}{10}$

Normal Distribution: Probability

Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

a. $p(x < 19.2)$

b. $p(15 < x)$

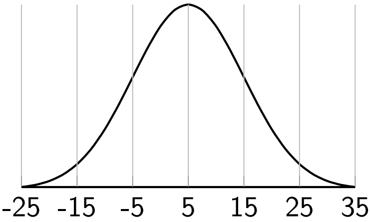
c. $p(11 < x < 15)$

Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.



a. $p(x < 19.2) = p(z < 1.42) = 92.22\%$

b. $p(x < 15) = p(z < 1) = 84.13\%$

c. $p(x < 11) = p(z < .6) = 72.57\%$

Normal Distribution: Probability

Example 4.4.5.

A population is normally distributed with a mean of 5 and a standard deviation of 10. Find the following:

a. $p(x < 19.2)$

b. $p(15 < x)$

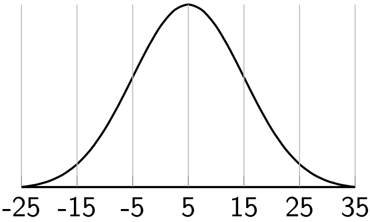
c. $p(11 < x < 15)$

Convert a Normal to a Z

If x is in a normal distribution then it is

$$z = \frac{x - \mu}{\sigma}$$

standard deviations from the mean.



a. $p(x < 19.2) = p(z < 1.42) = 92.22\%$

b. $p(x < 15) = p(z < 1) = 84.13\%$

c. $p(x < 11) = p(z < .6) = 72.57\%$

$p(11 < x < 15)$
$= p(.6 < z < 1)$
$= 84.13\% - 72.57\% = 11.56\%$

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
−3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
−3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
−3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
−3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
−3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
−2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
−2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
−2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
−2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
−2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
−2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
−2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
−2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
−2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
−2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
−1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
−1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
−1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
−1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
−1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
−1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
−1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
−1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
−1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
−1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
−0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
−0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
−0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
−0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
−0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
−0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
−0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
−0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
−0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
−0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.5000	.5040	.5080	.5120	.5160	.5199	.5239	.5279	.5319	.5359
0.1	.5398	.5438	.5478	.5517	.5557	.5596	.5636	.5675	.5714	.5753
0.2	.5793	.5832	.5871	.5910	.5948	.5987	.6026	.6064	.6103	.6141
0.3	.6179	.6217	.6255	.6293	.6331	.6368	.6406	.6443	.6480	.6517
0.4	.6554	.6591	.6628	.6664	.6700	.6736	.6772	.6808	.6844	.6879
0.5	.6915	.6950	.6985	.7019	.7054	.7088	.7123	.7157	.7190	.7224
0.6	.7257	.7291	.7324	.7357	.7389	.7422	.7454	.7486	.7517	.7549
0.7	.7580	.7611	.7642	.7673	.7704	.7734	.7764	.7794	.7823	.7852
0.8	.7881	.7910	.7939	.7967	.7995	.8023	.8051	.8078	.8106	.8133
0.9	.8159	.8186	.8212	.8238	.8264	.8289	.8315	.8340	.8365	.8389
1.0	.8413	.8438	.8461	.8485	.8508	.8531	.8554	.8577	.8599	.8621
1.1	.8643	.8665	.8686	.8708	.8729	.8749	.8770	.8790	.8810	.8830
1.2	.8849	.8869	.8888	.8907	.8925	.8944	.8962	.8980	.8997	.9015
1.3	.9032	.9049	.9066	.9082	.9099	.9115	.9131	.9147	.9162	.9177
1.4	.9192	.9207	.9222	.9236	.9251	.9265	.9279	.9292	.9306	.9319
1.5	.9332	.9345	.9357	.9370	.9382	.9394	.9406	.9418	.9429	.9441
1.6	.9452	.9463	.9474	.9484	.9495	.9505	.9515	.9525	.9535	.9545
1.7	.9554	.9564	.9573	.9582	.9591	.9599	.9608	.9616	.9625	.9633
1.8	.9641	.9649	.9656	.9664	.9671	.9678	.9686	.9693	.9699	.9706
1.9	.9713	.9719	.9726	.9732	.9738	.9744	.9750	.9756	.9761	.9767
2.0	.9772	.9778	.9783	.9788	.9793	.9798	.9803	.9808	.9812	.9817
2.1	.9821	.9826	.9830	.9834	.9838	.9842	.9846	.9850	.9854	.9857
2.2	.9861	.9864	.9868	.9871	.9875	.9878	.9881	.9884	.9887	.9890
2.3	.9893	.9896	.9898	.9901	.9904	.9906	.9909	.9911	.9913	.9916
2.4	.9918	.9920	.9922	.9925	.9927	.9929	.9931	.9932	.9934	.9936
2.5	.9938	.9940	.9941	.9943	.9945	.9946	.9948	.9949	.9951	.9952
2.6	.9953	.9955	.9956	.9957	.9959	.9960	.9961	.9962	.9963	.9964
2.7	.9965	.9966	.9967	.9968	.9969	.9970	.9971	.9972	.9973	.9974
2.8	.9974	.9975	.9976	.9977	.9977	.9978	.9979	.9979	.9980	.9981
2.9	.9981	.9982	.9982	.9983	.9984	.9984	.9985	.9985	.9986	.9986
3.0	.9987	.9987	.9987	.9988	.9988	.9989	.9989	.9989	.9990	.9990
3.1	.9990	.9991	.9991	.9991	.9992	.9992	.9992	.9992	.9993	.9993
3.2	.9993	.9993	.9994	.9994	.9994	.9994	.9994	.9995	.9995	.9995
3.3	.9995	.9995	.9995	.9996	.9996	.9996	.9996	.9996	.9996	.9997
3.4	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9997	.9998